Blur Detection and Smart Meter Reading System

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team

Project Title:		
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Project Overview

Punjab State Power Corporation Limited (PSPCL) relies on field officers to periodically collect electricity meter readings from consumers to generate accurate bills and monitor electricity consumption. During field visits, meter images are captured, and readings are recorded for billing and operational monitoring. The accuracy of this process is essential for reliable revenue management and effective detection of irregular electricity usage. However, the existing meter reading workflow is affected by several challenges, including unclear meter images, manual recording errors, and limited mechanisms for verifying the authenticity of submitted readings.

Under real-world field conditions, meter images are often captured in uncontrolled environments involving poor lighting, camera movement, and varying meter placements. These factors may result in blurred or unclear images, which reduce the reliability of automated reading extraction and may lead to inaccurate billing records. Furthermore, electricity distribution systems remain vulnerable to fraudulent activities such as meter tampering, abnormal consumption patterns, or intentional manipulation of readings. Because existing systems rely largely on manual verification and delayed investigations, the timely identification of suspicious cases becomes difficult.

To address these challenges, an AI-assisted smart meter reading verification system is proposed. The system incorporates a mobile application for capturing meter images, an image quality validation module for detecting blur, and Optical Character Recognition (OCR) techniques for automated extraction of meter readings. An app with a backend processing system and an administrative dashboard is also integrated to enable monitoring and validation of submitted data. Suspicious readings can be automatically flagged and escalated to the Local Complaint Resolution (LCR) team for further verification. Through the integration of automated image validation, intelligent data extraction, and centralized monitoring, the proposed system aims to improve the accuracy, transparency, and reliability of PSPCL’s meter reading operations.

Problem Statement

The integrity of PSPCL's billing ecosystem is currently compromised by three primary vulnerabilities:

Image Quality and Extraction Failures:

Uncontrolled field conditions (e.g., poor lighting, motion) frequently produce blurred images. Without real-time quality validation, these degraded images cause Optical Character Recognition (OCR) failures, leading to inaccurate billing and forced manual intervention.

Meter Reading Fraud and Manipulation:

A heavy reliance on manual data entry enables intentional false reporting and meter tampering. The lack of automated "Proof of Presence" or on-site verification makes it difficult to detect these fraudulent activities early.

Inefficient Verification and Investigation:

The absence of automated consistency checks means anomalies are typically only identified reactively—often after consumers raise billing disputes. Consequently, the Local Complaint Resolution (LCR) team cannot proactively review suspicious readings at the point of capture.

These deficiencies directly result in billing inaccuracies, delayed fraud detection, and significant revenue leakage. An intelligent, multi-tiered verification system is essential to automatically validate image quality, ensure accurate OCR extraction, and proactively flag suspicious submissions.

Need Analysis

Electricity distribution companies (DISCOMs) in India, particularly those operating in geographically diverse and remote rural areas, face significant operational and financial challenges with both traditional manual recording and existing smart metering systems.

Current Challenges

Inaccuracies and Vulnerabilities in Manual Reading: Relying on purely manual data entry introduces a high probability of human error and creates avenues for record manipulation. Without systematic image verification, DISCOMs suffer from inaccurate billing, which directly contributes to Aggregate Technical and Commercial (AT&C) losses and increases the frequency of consumer disputes [1].

Connectivity Constraints in "Zero-Network" Zones: Traditional internet-dependent mobile applications fail frequently in rural and remote regions characterized by intermittent or non-existent cellular coverage [2]. This reliance on continuous connectivity leads to incomplete data transmission, delayed billing cycles, and administrative inefficiencies due to the disruption of real-time data flow between field workers and centralized servers.

Technical and Economic Limitations of Existing Smart Systems: While Advanced Metering Infrastructure (AMI) and IoT-based smart meters offer automated solutions, their widespread deployment across all operational regions is often hindered by high capital costs, legacy infrastructure bottlenecks, and vulnerability to network outages [3]. Furthermore, current transitionary solutions (like standard mobile data-entry apps) lack edge-computing capabilities. They commonly fail to provide real-time image quality assessment (such as blur detection), OCR confidence scoring, secure offline data caching, and structured escalation workflows for reading discrepancies [4]. Consequently, blurred or poorly captured meter images result in corrupted Optical Character Recognition (OCR) extraction, ultimately negating the efficiency of automated billing.

Literature Survey

Optical Character Recognition for Automated Meter Reading

Optical Character Recognition (OCR) is commonly used for automated data extraction in areas like document digitization, license plate recognition, and utility meter reading [2]. Traditional OCR engines such as Tesseract depend on image preprocessing steps like binarization, segmentation, and noise reduction. These methods work well on clear, structured images but struggle a lot in real-world field conditions with blur, noise, or changing light [1].

Recent advances in deep learning have created stronger OCR systems using Convolutional Neural Networks (CNNs) and sequence models [3]. These often combine YOLO for detection and CRNN for recognition, handling issues like skewness and cluttered backgrounds better. Peng et al. (2023) introduced a YOLO-based system for real-time digital meter reading in natural scenes with varying light, scale, and angles. It uses a corner point detection module with attention mechanisms, dynamic loss weighting, and data augmentation (rotation and flipping). Detected corners allow perspective correction, followed by end-to-end recognition. On a custom dataset of 1,255 challenging images, it reached 99.6% accuracy with just 8.6 ms inference time, beating earlier methods [7]. Khan et al. (2024) built a low-cost pipeline with YOLOv8 for distant meter detection and fine-tuned Paddle OCR. On 8,044 images of electricity, gas, and water meters, it achieved 0.995 mAP for detection, 96.92% recognition accuracy (70% better than baseline), and 97.8% end-to-end at 6 frames per second—great for mobile use [8].

For analog dial meters, Salomon et al. (2020) released the UFPR-ADMR dataset with 2,000 real-world images facing blur, reflections, dirt, and uneven lighting. Their Faster R-CNN with ResNeXt-101 got 100% F1-score for dial detection and 93.6% for recognition, giving 75.25% full meter accuracy and clearly outperforming traditional Hough Circle methods [9]. Still, most deep learning OCR models assume fairly clear images and do not fully handle severe blur right at capture [2,7].

Blur Detection Techniques in Image Quality Assessment

Blur detection is a key part of image quality assessment. Traditional techniques measure edge sharpness with methods like the Variance of the Laplacian, gradient analysis, and frequency domain filtering [4]. These are lightweight and good for real-time mobile processing, but they can struggle with uneven lighting or textured backgrounds [5].

More recent work uses machine learning and deep learning for blur classification. Lin et al. (2023) combined FFT-based frequency-domain detection (more stable than Laplacian) with an improved DeblurGANv2 network that includes spatial attention, AdaIN, and SiLU activations [10]. This preprocessing feeds a full meter-reading pipeline: Polygon-YOLOv5 for LED regions, perspective correction, YOLOv5s for digits, and CRNN for sequences. The system hits 98% reading accuracy with only 1% missing rate, and CRNN accuracy jumps from 86.4% (no preprocessing) to 98.8% in tough lighting and motion blur from robots. Hybrid classical-deep learning approaches perform better while staying suitable for on-device use.

Smart Utility Monitoring and Automated Meter Reading Systems

Many smart utility systems use mobile apps and IoT devices to automate meter reading and cut down on human work [6]. However, most focus only on recognition and transmission, skipping strong image quality checks on the worker's phone. This often leads to wrong readings or failures from poor field captures [1].

Adding quick image quality assessment before OCR greatly improves reliability, reduces errors, and gives instant feedback to retake photos [3]. New setups like YOLOv8 with Paddle OCR and corner-corrected YOLO show real-time performance is now possible on mobile devices [7,8].

Probabilistic and Uncertainty-Aware Approaches in Meter Data Analysis

AI in smart metering also uses probabilistic models to deal with uncertainty in consumption data. Roth et al. (2021) applied Bayesian Structural Time Series (BSTS) for probabilistic load forecasting on sub-hourly data from 120 apartments [11]. It uses spike-and-slab priors for feature selection (like temperature and submeter loads) and MCMC sampling to create full prediction intervals. BSTS beat traditional ARIMA in accuracy and supports segmentation, demand-response checks, and risk-aware operations—especially helpful when noisy OCR readings feed into later analysis.

Research Gap

The literature shows that current OCR-based meter reading solutions often miss a lightweight on-device step for blur detection and quality validation. Classical methods like Laplacian or FFT [4] and advanced models like YOLO, CRNN, and Paddle OCR [10] work well separately in controlled settings, but their full integration into one real-time mobile pipeline with uncertainty handling is still limited. Partial fixes exist from Lin et al. (FFT + DeblurGAN) and Peng et al. (corner correction) [7,10], yet few add immediate quality checks on workers' phones before recognition. Datasets also point to ongoing issues with combined blur, tilt, and lighting in mobile captures. We clearly need a complete end-to-end system with instant quality assessment, hybrid OCR, and probabilistic steps in one workflow to deliver reliable data in real field conditions.

Objectives

This capstone project aims to develop a secure, scalable, and intelligent meter reading system for PSPCL by achieving the following primary objectives:

1. Design an Intelligent Mobile Application

We will develop a secure app for field officers to efficiently capture and submit meter images. It will include local data storage and offline synchronization to ensure reliable operation in real-world, low-network conditions.

2. Integrate AI Image Validation and OCR

We will implement an AI pipeline using the Variance of Laplacian method for real-time blur detection. Additionally, we will integrate OCR to automatically extract meter readings and numbers, providing confidence scores for validation.

3. Develop a Secure Backend with Fraud Detection

A secure backend infrastructure will be built for authentication and data validation. It will detect abnormal consumption patterns, automatically flag suspicious cases, and escalate them to the Local Complaint Resolution (LCR) team.

4. Construct an Administrative Audit Dashboard

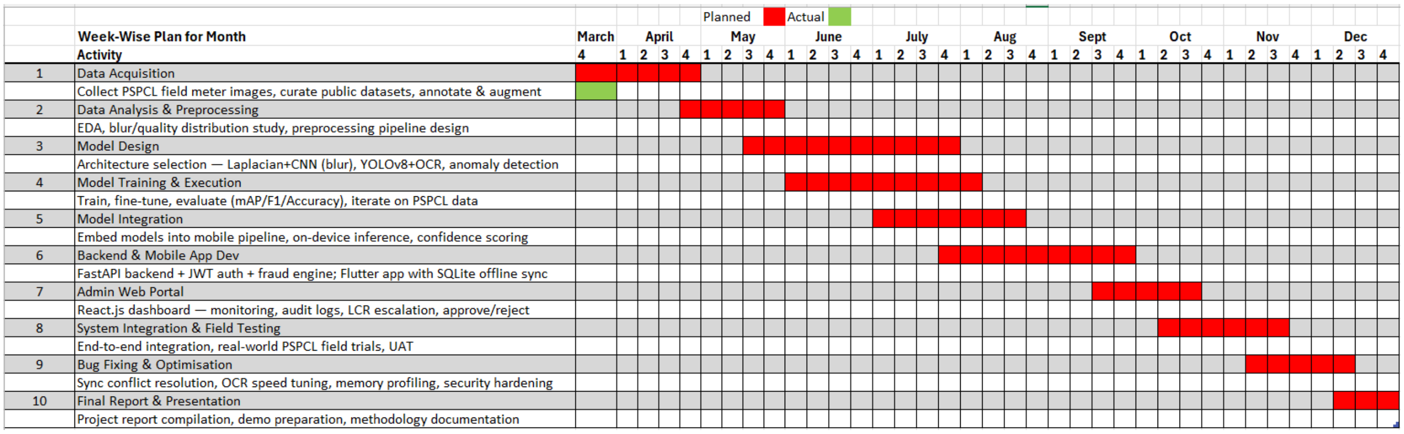
A web-based dashboard will enable PSPCL officials to monitor readings, verify images, and manage flagged cases. Administrators will be able to approve or reject submissions while maintaining detailed audit logs for full transparency.

Together, these objectives deliver a system that enhances reading accuracy, improves fraud detection, and streamlines electricity distribution monitoring.

Methodology

- You need to discuss the stepwise methodology of achieving the set objectives in this section. (Diagram along with stepwise description is a good tool to explain the methodology)

Work Plan



Project Outcomes & Individual Roles

The successful execution of this capstone project will result in the following key deliverables:

Intelligent Mobile Meter Reading Application: A functional mobile application designed for field officers to capture meter images efficiently, featuring local data storage and synchronization to operate reliably under real-world, intermittent network conditions.

Real-Time Image Validation and OCR Module: An integrated processing pipeline utilizing the Variance of Laplacian method for instant blur detection, coupled with Optical Character Recognition (OCR) to automatically extract meter numbers and consumption readings along with validation confidence scores.

Secure Backend and Fraud Detection System: A centralized backend infrastructure that manages authentication, analyzes submitted data for abnormal consumption patterns, and automatically escalates suspicious cases to the Local Complaint Resolution (LCR) team.

Administrative Audit Dashboard: A comprehensive web-based portal enabling officials to monitor submitted data, review flagged cases, approve or reject submissions, and maintain detailed audit logs for transparency and accountability.

Table 1: Technical roles and responsibilities of individual team members.

Team Member	Technical Role and Responsibilities
Ananya Singh	Full-stack development is led, including backend integration and the design of the PostgreSQL database schema.
Nishtha Gupta	The Flutter mobile application is developed, focusing on UI/UX design, camera integration, and offline features.
Arjun Angirus	Data handling processes are managed, along with backend support and the implementation of system analytics.
Shrey Goyal	Machine Learning model development is conducted, followed by the deployment of services to the cloud environment.
Madhav Arora	Core machine learning algorithms are implemented and optimized to ensure high accuracy on edge devices.

Course Subjects

The course subjects that have been taught during the program and have been applied in the successful execution of this project are as follows:

Software Engineering - Used for requirement analysis of the PSPCL system, designing the OCR workflow, modular development of the application and systematic testing of the complete solution.

Artificial Intelligence - Applied for enabling intelligent automation in meter reading interpretation and decision-making processes within the OCR-based system.

Machine Learning - Utilized for training models to recognize patterns in meter images and improve prediction/classification accuracy for extracted readings.

Deep Learning - Used for implementing neural network-based OCR techniques and image-based feature learning to enhance recognition performance.

Predictive Analysis and Statistics - Applied for analysing meter data trends, validating model performance using statistical metrics and improving forecasting reliability

Database Management Systems - Used for storing extracted readings, user/app data and maintaining structured datasets required for model training and dashboard visualization.

Image Processing - Applied for preprocessing meter images such as blur detection, noise removal, segmentation and feature extraction before OCR model execution.

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Formatting Guidelines

- Project Report Type: Transparencies and tape bound
- Number of Copies: 1 per Project group (Max pages 15)
- Running text should be justified, figures and tables center aligned, no space before full stop etc.
- Use **passive voice** in text.
- Paper Size (orientation): A4 (portrait)
- Margins: 1” top / bottom / right and 1.5” left
- Font Type: Times New Roman
- Font Size: 16 bold for Section names, 14 bold for headings and 12 for normal text
- Line Spacing: 1.5 throughout
- Page Numbering: Bottom center of page in the format – Page 1 of N
- All table and figure captions in size 10 sentence case, table captions on top and figure captions below the figure.
- All figures and tables quoted in the text with explanation.
- No figures and equations should be copied. Please use **smartdraw/ visio for figures/FIGMA and Mathtype** for equations.
- References (The listing of references should be typed 2 spaces below the heading “REFERENCES” in alphabetical order in single spacing left – justified. It should be numbered consecutively (in square [ ] brackets, throughout the text and should be collected together in the reference list at the end of the report. The references should be numbered in the order they are used in the text. The name of the author/authors should be immediately followed by the year and other details). References should not be cited from Blogs, Twitter etc. but should refer to good Journal or Conference papers. Typical examples of the references are given below: